

UK Maths Trust

Junior Mathematical Challenge

Wednesday 29 April 2026

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MARKETS

England & Wales: Year 8 or below | Scotland: S2 or below | Northern Ireland: Year 9 or below

Instructions

1. Do not open the paper until the invigilator tells you to do so.
2. Time allowed: **60 minutes**. No answers, or personal details, may be entered after the allowed time is over.
3. The use of blank paper for rough working is allowed; **squared paper, calculators and measuring instruments are forbidden**.
4. **Use a B or an HB non-propelling pencil**. Mark at most one of the options, A, B, C, D, or E, on the Answer Sheet for each question. Do not mark more than one option.
5. **Do not expect to finish the whole paper in the time allowed**. The questions in this paper have been arranged in approximate order of difficulty with the harder questions towards the end. You are not expected to complete all the questions during the time. You should bear this in mind when deciding which questions to tackle.
6. **Scoring rules:**
5 marks are awarded for each correct answer to Questions 1-15;
6 marks are awarded for each correct answer to Questions 16-25.
There is no penalty for giving an incorrect answer.
7. **Your Answer Sheet will be read by a machine**. Do not write or doodle on the sheet except to mark your chosen options. The machine will read all markings even if they are in the wrong places. If you mark the sheet in the wrong place, doodle, or leave bits of eraser stuck to the page, the machine will interpret the mark in its own way.
8. **The questions on this paper are designed to challenge you to think, not to guess**. You will gain more marks, and more satisfaction, by doing one question carefully than by guessing lots of answers. This paper is about solving interesting problems, not about lucky guessing.
9. To accommodate candidates sitting at other times, please do not discuss the paper on the internet until **08:00 BST on Friday 1st May**, when the solutions video will be released at ukmt.org.uk/competition-papers. Candidates in time zones more than 2 hours ahead of GMT must sit the paper on Thursday 30th April (as defined locally).

Enquiries about the Junior Mathematical Challenge should be sent to:

challenges@ukmt.org.uk www.ukmt.org.uk

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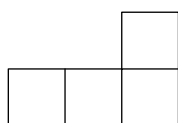
1. Which of the following calculations gives the largest result?

A $2 \times 0 + 26$ B $2 + 0 + 2 \times 6$ C $20 + 26$ D $20 \times 2 - 6$ E $20 + 2 \times 6$

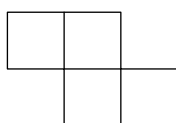
2. Each figure below consists of four squares of side-length 1 cm.

In which figure is the length of the perimeter different from that of the other four?

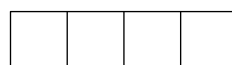
A



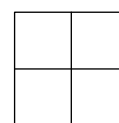
B



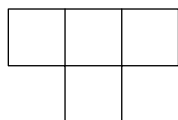
C



D



E



3. What is the value of $\frac{4 - 3 - 2 - 1}{8 - 7 - 6 - 5}$?

A 0

B $\frac{1}{5}$

C $\frac{1}{4}$

D $\frac{1}{3}$

E $\frac{1}{2}$

4. Barry thinks of a number. Half of the number, plus a third of the number, plus a quarter of the number is 52. What number is Barry thinking of?

A 24

B 30

C 36

D 42

E 48

5. The Salisbury Cathedral clock is thought to be the world's oldest working clock, dating from about 1386. It was restored in 1956. Since its restoration, the clock strikes once on the hour every hour. Which of these is the best approximation for the number of times it has struck since its restoration?

A 25 000

B 200 000

C 300 000

D 600 000

E 1 200 000

6. Ada likes adding numbers! On Monday she added all the three-digit numbers each of whose digits is either 0 or 1. On Tuesday she added all the two-digit numbers each of whose digits is either 2 or 3. On Wednesday she added her answers from Monday and Tuesday. What was the result of Ada's final addition?

A 310

B 421

C 532

D 643

E 754

7. Sam is asked to write 2026 as the sum of a list of five different primes. How many possibilities are there for the smallest prime in Sam's list?

A one

B two

C three

D four

E more than four

8. Rectangle $PQRS$ has perimeter 30 cm.

The area of square $PUTX$ is 9 cm^2 .

What is the perimeter of rectangle $TVRW$?

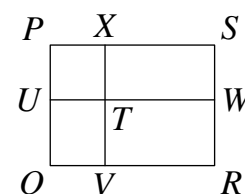
A 12 cm

B 15 cm

C 18 cm

D 21 cm

E 24 cm



9. Carl counts in a curious way! He starts at 1, but then always misses out multiples of three and multiples of five. What number would Carl say last when he counts 15 items?

A 22

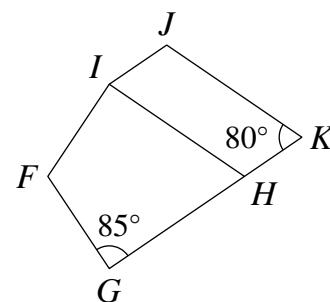
B 23

C 26

D 28

E 29

10. The diagram shows kite $FGHI$, in which $IF = FG$ and $GH = HI$, and parallelogram $HIJK$. Points G, H and K lie on a straight line. Angle JKH is 80° and angle FGH is 85° . What is the size of angle IFG ?

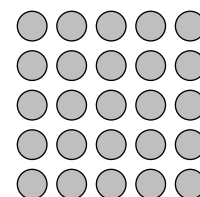


A 100° B 110° C 120° D 125° E 150°

11. In how many ways can all the digits 1, 2, 3, 4, 5, 6, 7, 8 and 9 be placed in a 3×3 grid so that the sum of the numbers in each row is even and the sum of the numbers in each column is odd?

A 0 B 1 C 2 D 3 E more than 3

12. John is given a 5×5 array of biscuits as shown. He is told that he can eat these one by one, provided that each time the remaining array has at least one line of symmetry. What is the largest number of biscuits that John can eat?

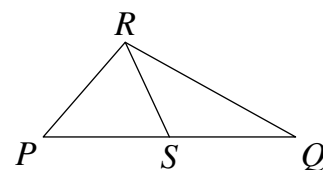


A 5 B 9 C 13 D 17 E 25

13. The letters p, q and r represent digits. The sum of the two-digit numbers ' pq ' and ' pr ' is 75. What is the value of $p + q + r$?

A 12 B 14 C 15 D 16 E 18

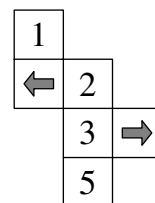
14. In the triangle shown, S is a point on the base PQ . The perimeter of triangle PQR is 80 cm. The perimeter of triangle RPS is 44 cm. The perimeter of triangle RSQ is 60 cm. What is the length of RS ?



A 20 cm B 18 cm C 16 cm D 14 cm E 12 cm

15. The diagram shows the net of a cube. When it is folded to create a cube, each arrow points to an adjacent face. What is the sum of the two numbers on those faces?

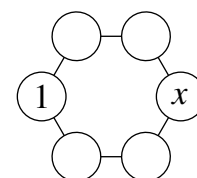
A 3 B 4 C 5 D 6 E 7



16. The numbers 1, 2, 3, 4, 5, and 6 are to be placed in the six circles shown so that:

- (i) there is one number in each circle, and
- (ii) the sum of the numbers in each pair of adjacent circles is a prime number.

The number 1 has already been placed. Which number should go in the circle marked x ?



A 2 B 3 C 4 D 5 E 6

17. Hanlin writes down three 1s and three 0s, according to the following rule: while writing down the numbers, he must always have written at least as many 1s as he has written 0s. How many such numbers can he form from the six digits?

A 4 B 5 C 6 D 7 E 8

18. Five children, Amy, Bob, Cat, Dan and Eve each make a statement about the number of pencils they have in their pencil case.

Amy: "I have more than Bob."

Bob: "I have more than Cat and more than Dan."

Cat: "I have more than Dan."

Dan: "I have more than Eve and more than Amy."

Eve: "I have more than Amy."

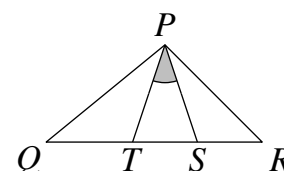
What is the largest number of these statements that can be true?

- A 1 B 2 C 3 D 4 E 5

19. In the triangle QPR , $\angle QPR = 100^\circ$, $QP = QS$ and $RP = RT$.

What is the size of $\angle TPS$?

- A 30° B 35° C 40° D 45°
E it depends on the size of $\angle RQP$.



20. In Kush's Cake shop, there are chocolate cakes, cherry cakes, cream cakes and coffee cakes. For every seven chocolate cakes, there are six cherry cakes. For every five cherry cakes, there are four cream cakes. For every three cream cakes, there are two coffee cakes. Which of these could be the total number of cakes in Kush's shop?

- A 60 B 150 C 240 D 330 E 420

21. What is the smallest number of rectangles, each measuring 5 cm by 30 cm, which can be fitted together without gaps or overlapping to form a rectangle whose side-lengths are in the ratio 4 : 5?

- A 15 B 30 C 45 D 60 E 75

22. In Poppy's pond there are frogs and snakes. Four-fifths of the frogs are green frogs and six-sevenths of the snakes are black snakes. Which of the following could *not* be the total number of creatures in Poppy's pond?

- A 22 B 23 C 24 D 26 E 27

23. Five different numbers are written in ascending order.

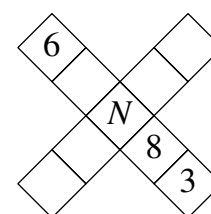
The mean of the first four numbers is 5. The mean of the last four numbers is 7.

What is the range of the numbers?

- A 2 B 4 C 6 D 7 E 8

24. The diagram shows nine squares arranged so that they form two lines, each consisting of five squares.

The squares are to be filled with the positive integers from 1 to 9 inclusive so that the totals of the five numbers in each line are equal. Three of the numbers have already been placed. What is the sum of the different values which the central number N can take?



- A 17 B 21 C 22 D 25 E 28

25. Four runners entered a race. We are told that at most two tied at the finish line (and were therefore given the same finishing position). How many possible finishing orders were there for the race?

- A 30 B 60 C 75 D 81 E 84